

Chemistry

CYL1010



Google Class code: **izdaj53f**

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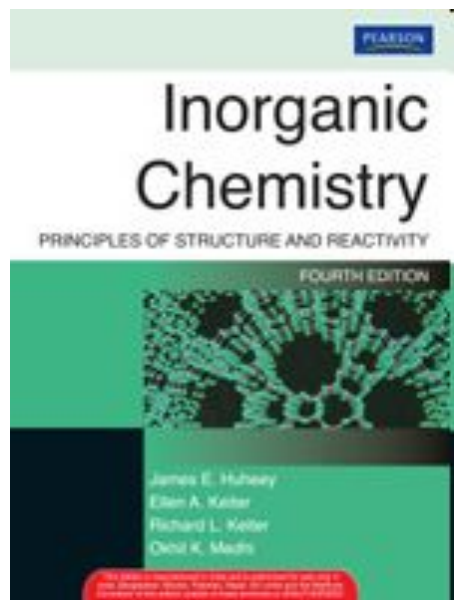
Contents

Title	Chemistry	Number	CYL1010
Department	Chemistry	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. 1 st Year	Type	Compulsory
Prerequisite	Nil		
Objectives 1. Relate the fundamental principles of chemistry with practical problems encountered for engineers and place the emphasis on problem-solving 2. Provide knowledge of engineering materials metals, composites, polymers, cement, steel, fuels, batteries and drugs. Learning Outcomes 1. Understanding the behavior of matter and materials using fundamental knowledge of their nature. 2. Understanding the basic concepts and implementation of the same in advanced and interdisciplinary areas. 3. Predict potential complications from combining various chemicals or metals in an engineering setting. Contents <i>Atomic and Molecular Structure [6 Lectures]:</i> particle in a box, angular momentum, wavefunction, multielectron atoms, homonuclear and heteronuclear diatomic molecules, non-covalent interactions <i>Solid State Chemistry [6 Lectures]:</i> Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties. <i>Surface Chemistry[3 Lectures]:</i> Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension. <i>Electrochemical Systems [5 Lectures]:</i> Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion <i>Basic Spectroscopy [7 Lectures]:</i> Selection Rules, Fundamentals FTIR, Raman , UV-visible and Fluorescence spectroscopy <i>Transition Metal Chemistry [5 Lectures]:</i> Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry <i>Organic Chemistry [8 Lectures]:</i> Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies. Textbooks 1. Silberberg, M., <i>Chemistry: The Molecular Nature of Matter and Change</i>, 6th Edition, McGraw Hill Education. 2. West, A.R., (2015), <i>Solid State Chemistry and Its Applications</i>, 2nd edition, John Wiley & Sons. 3. J. D. Lee, <i>Concise Inorganic Chemistry</i>, 5th Edition. 4. Atkins, <i>Physical Chemistry</i>, 11th Edition. Reference Books 1. McMurry, J. E. & Fay, R. C. <i>Chemistry</i>, 5th Edition, Pearson. 2. Jonathan Clayden, Nick Greeves, Stuart Warren, <i>Organic Chemistry</i>, 2nd Edition. 3. F. Albert Cotton, Geoffrey Wilkinson, <i>Basic Inorganic Chemistry</i>, Wiley Student Edition. Self Learning Material 1. https://nptel.ac.in/downloads/122101001/			

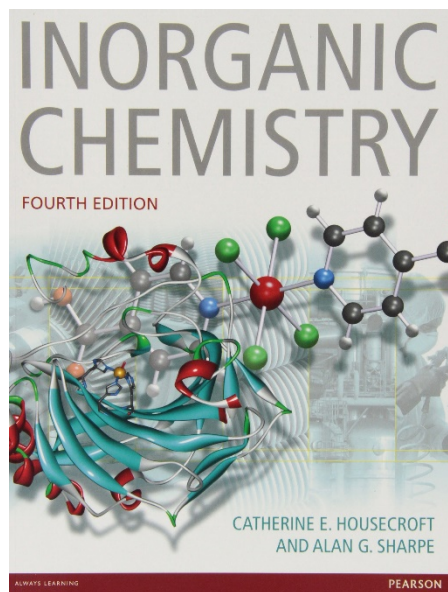
Contents

- **Solid State Chemistry [6 Lectures]:** Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties.
- **Surface Chemistry[3 Lectures]:** Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension.
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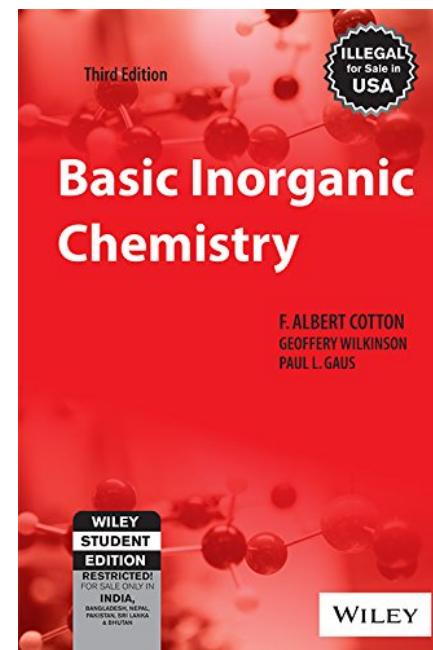
Reference Books



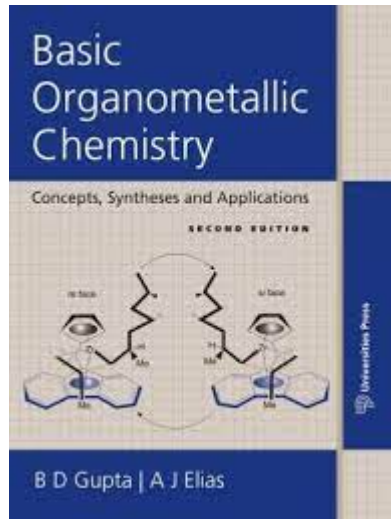
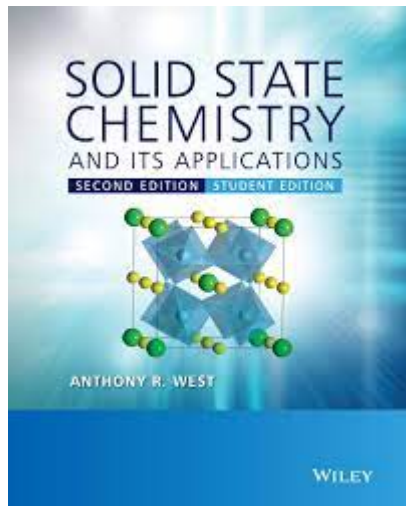
J. E. Huheey
Keiter & Keiter



Catherine E. Housecroft
and Alan G. Sharpe



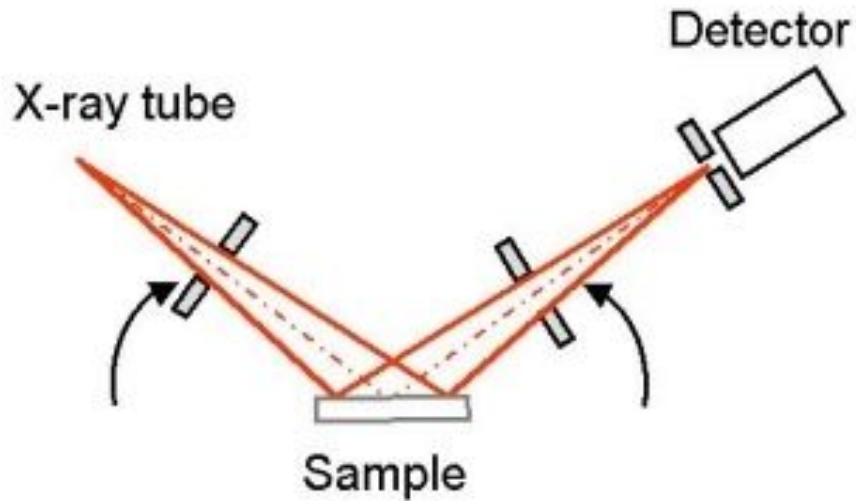
F. Albert Cotton,
Geoffrey Wilkinson



Course Grading Scheme

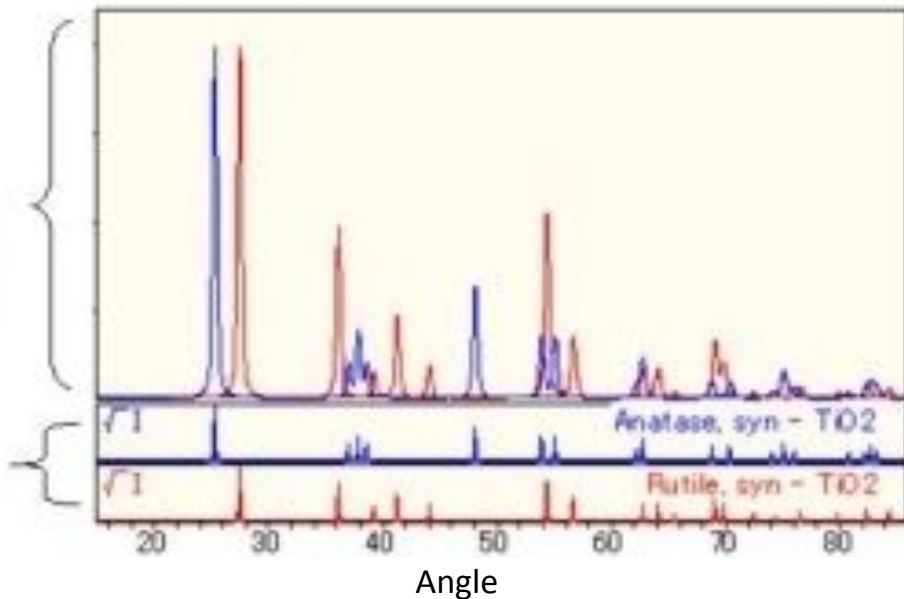
Quiz 1	10%
Quiz 2	10%
Assignment	5 %
Interaction	5%
Minor	30%
Major	40%
TOTAL	100 %

X-ray Diffraction (XRD)



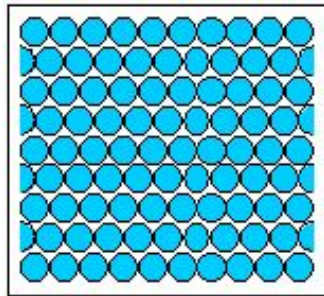
What Information Do We Get or Can We Get From Powder X-ray Diffraction

- ★ Lattice parameters
- ★ Phase identity
- ★ Phase purity
- ★ Crystallinity
- ★ Crystal structure
- ★ Percent phase composition

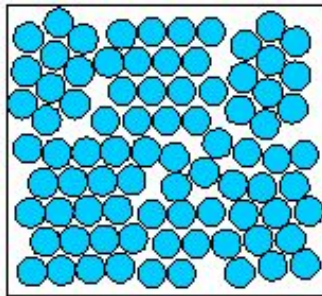


single crystal vs polycrystalline vs Amorphous solid

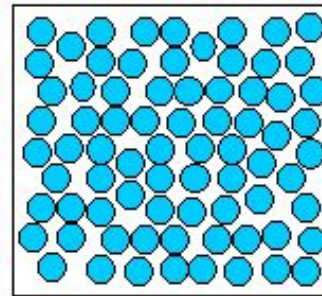
This is explained in terms of the atomic scale periodicity: **single crystals are periodic across their entire volume**; **polycrystals are periodic across individual grains**; amorphous solids have little to no periodicity at all. The different atomic structures can have effects on the macroscopic properties.



Single crystal
Periodic across the
whole volume.



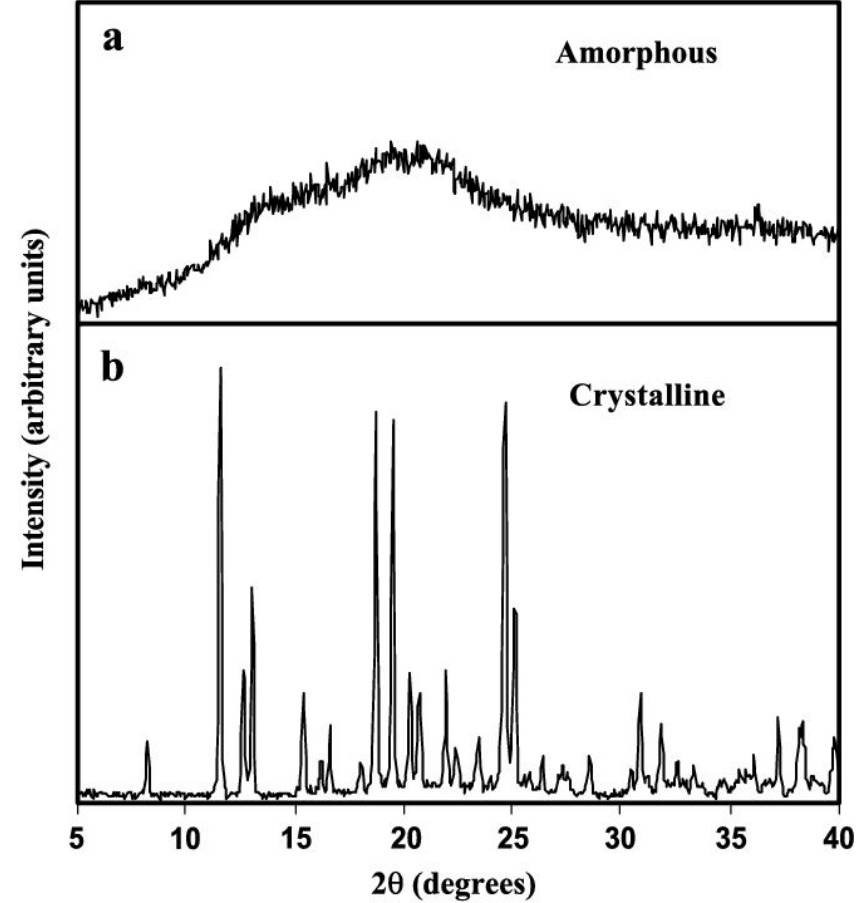
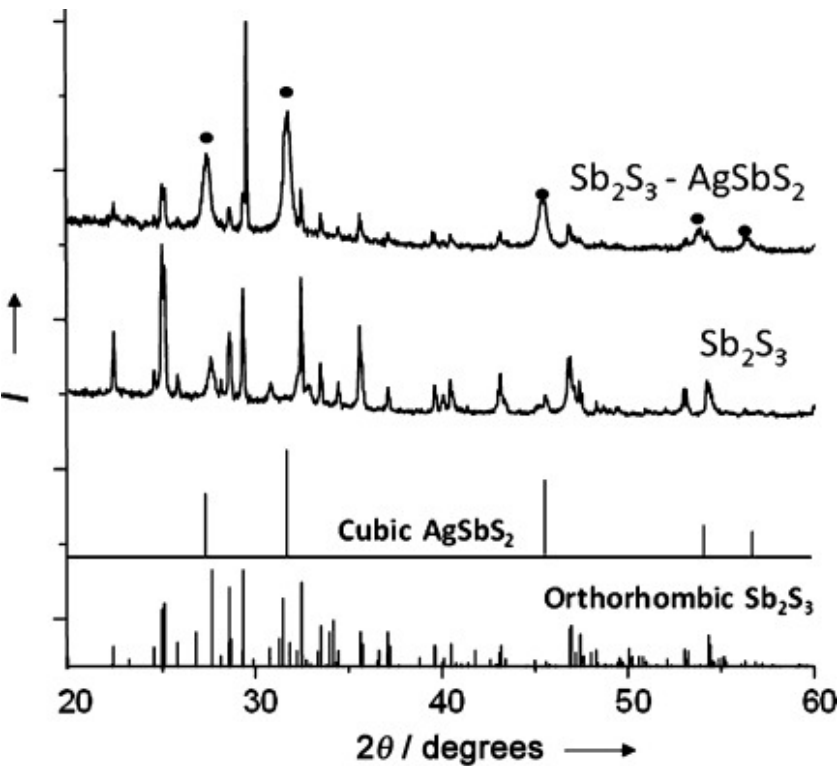
Polycrystal
Periodic across
each grain.



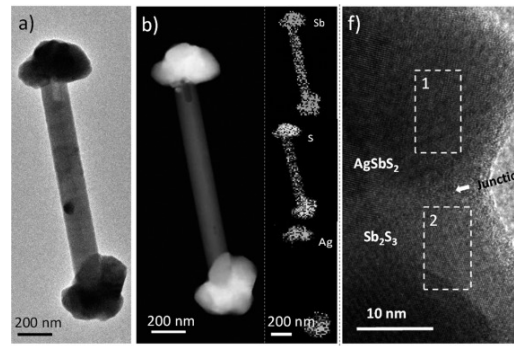
Amorphous solid
Not periodic.

- **X-ray diffraction (XRD)**

- Small angle XRD (SAXS)
- Wide angle XRD (WAXS)



Powder X-ray patterns of (a) amorphous, and (b) crystalline sucrose.



Important facts on amorphous solid

What is meant by an amorphous solid?

An amorphous solid is a type of matter (solid, liquid, gas) that lacks regular long-range order within its atomic structure. The lack of atomic-level order differentiates amorphous solids from crystalline solids, which have a regular and repeating arrangement of atoms. Another defining characteristic of amorphous solids is that they lack a regular geometric shape. While crystalline solids (for example, diamonds, sugar, salt, and snowflakes) commonly occur in geometric forms that reflect the shape and symmetry of their atomic-level order. The lack of atomic order also gives amorphous solids unique properties, such as the lack of a well-defined melting point, irregular fracture characteristics, poorly defined x-ray diffraction patterns, and isotropic properties (e.g., uniform mechanical strength, refractive index, and electrical and thermal conductivity).

What are the characteristics of an amorphous solid?

Amorphous solids lack an ordered atomic structure and therefore do not have well-defined melting points and do not form regular geometric shapes. The lack of atomic order and lack of a well-defined melting point give amorphous solids unique and useful characteristics such as the ability to be worked into useful shapes when heated (as used in glass blowing and the manufacture of bendable and flexible electronics). The unique characteristics of amorphous solids have led to technological advancements in materials science (e.g., using amorphous materials to form resistant coatings) and drug development (e.g., using amorphous materials to enhance the bioavailability of pharmaceuticals).

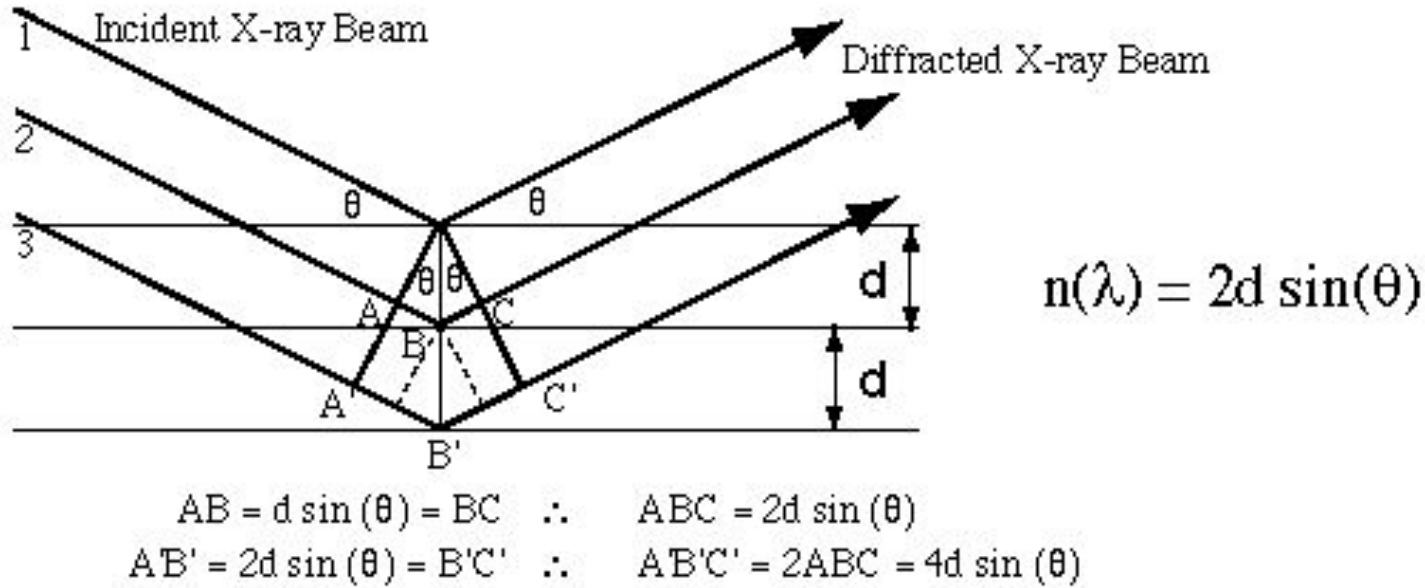
The most important characteristics of amorphous solids are that they: (1) **do not have long-range, regular atomic order**, (2) **form irregular shapes**, (3) **soften over a range of temperatures depending on their composition and thermal history**, (4) **fracture in irregular, unpredictable patterns** (e.g., conchoidal fracture), (5) **do not produce characteristic, sharp, diffraction patterns due to the lack of regular, symmetric crystal planes**, (6) **tend to deform in irregular, unpredictable ways due to the lack of regular internal structure**, (7) **are isotropic**. That is, their physical properties such as electrical and thermal conductivity, refractive index, and mechanical strength are uniform within the material due to the lack of a regular atomic structure.

Interaction of X-ray with Matter (sample)

- Fluorescence
- Ionization
- **Diffraction**

Electron beam Diffraction

Brag's Law



These *reflections* occur only under certain conditions which satisfy the equation:

$$n(\lambda) = 2d \sin \theta$$

The above equation is also known as Bragg's equation. Here n is an integer (1,2,3,...n), λ is the wavelength, d is the distance between the atomic planes, and θ is the angle of incidence of the X-ray beams. An X-ray beam takes a longer (but parallel) path because it 'reflects' off an adjacent atomic plane. This path length difference must equal an integer value of the wavelength of the incident X-ray beams for the constructive interference to occur such that a reinforced diffracted beam is produced

The Bragg's Law Formula

$$n\lambda = 2d \sin\theta$$

Where:

- n = order of reflection (1, 2, 3...)
- λ = wavelength of the X-ray
- d = distance between crystal planes
- θ = angle of incidence where diffraction occurs

Key Point: Diffraction peaks only happen at specific angles!

How does the
process happen?

1. X-rays strike the crystal at angle θ
2. Atoms are arranged in repeating planes.
3. They reflect off planes separated by d
 - Scattered X-rays interfere with each other
4. Waves interfere:
 - If path difference = $n\lambda$, they reinforce
 - Otherwise, they cancel
 - Constructive interference produces strong reflections
 - Bragg's law predicts the angles where this happens
5. By measuring θ , we can find d

Example problem:

If $\lambda = 1.54 \text{ \AA}$, $\theta = 30^\circ$, and $n = 1$, find d .

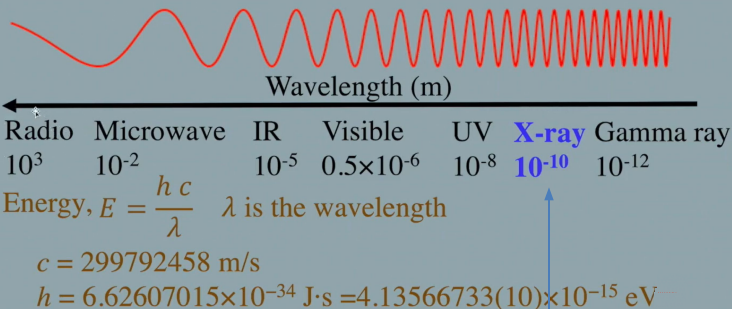
What, why and how of XRD

What is X-ray?

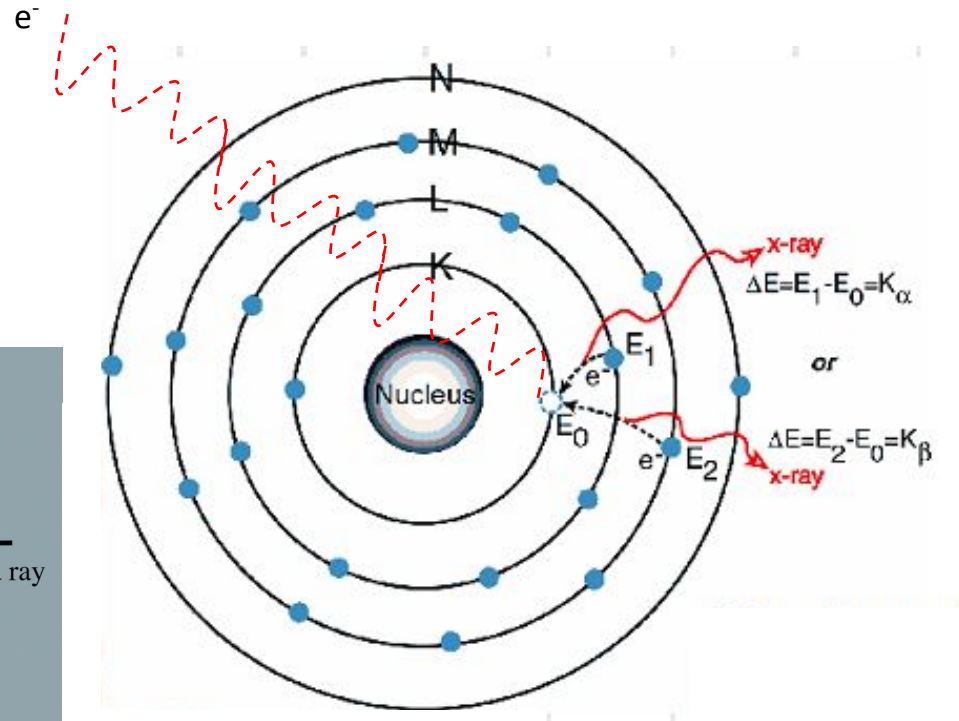
X-ray is electromagnetic radiation.

X-rays are produced by bombarding high energy electrons on any heavy metal.

Electromagnetic wave



X-ray production process

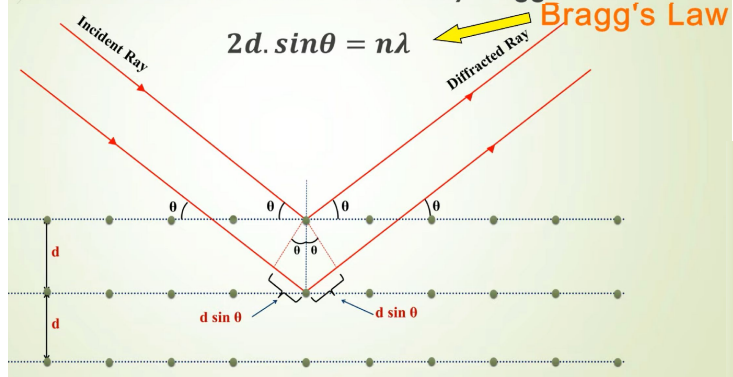


An electron jumps from L or M shell to fill the vacancy, generating X-rays

Why is it called X-ray Diffraction ?

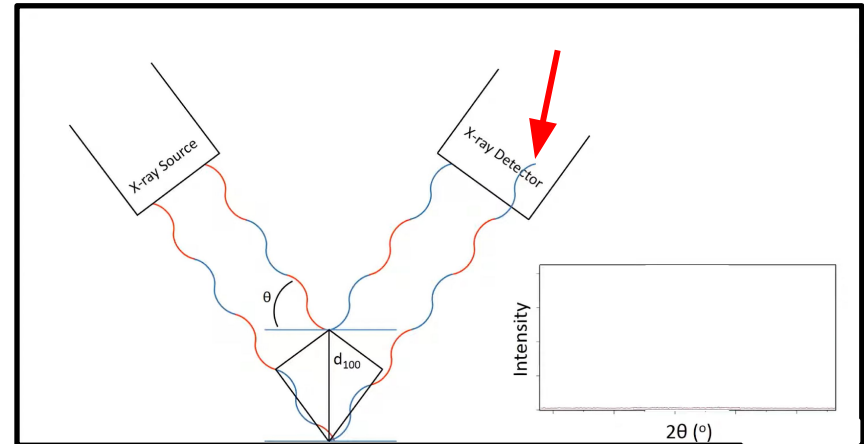
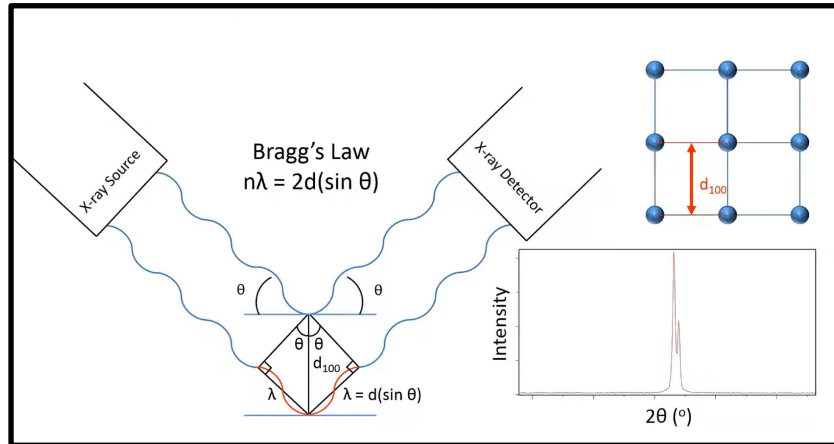
How XRD?

When atomic planes act as diffraction slits, then the constructive interference is defined by Bragg's Law—



Wave Interacting with a Solid or Crystalline Material

1. Scattered beams interfere constructively in some directions
2. Random arrangements cause beams to destructively interfere



It's called **X-ray diffraction** because X-rays are diffracted (bent and scattered) by the atomic planes inside a crystal, creating patterns that reveal the arrangement of atoms.

How is it called X-ray Diffraction ?

Key points about the relation between λ and d :

1. For diffraction to occur, λ must be comparable to d .

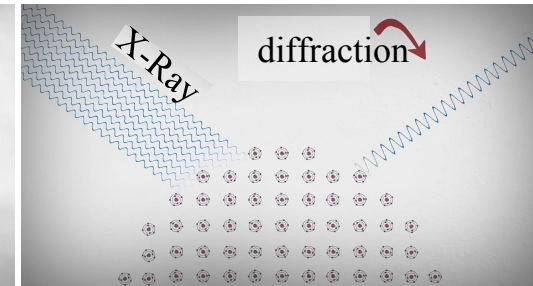
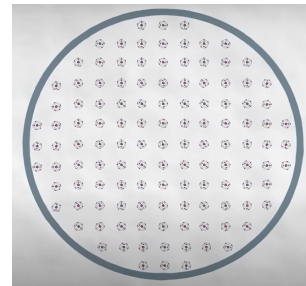
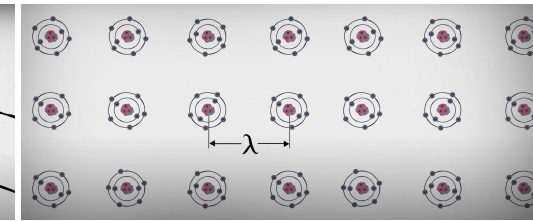
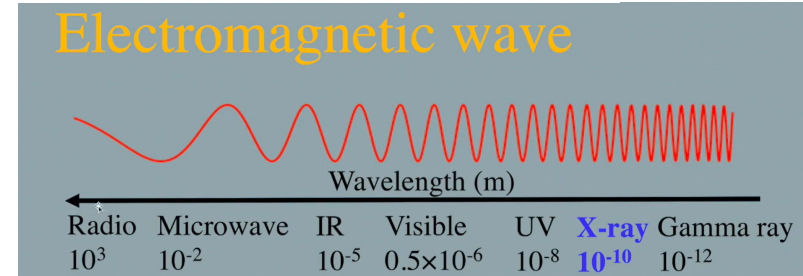
1. If λ is much larger than $d \rightarrow$ no clear diffraction pattern
2. If λ is much smaller $\rightarrow$ diffraction effects are too weak

2. The angle θ depends on the ratio of λ and d .

1. For a given λ , only certain angles θ will satisfy the equation and produce diffraction peaks.

3. The order n allows multiple solutions.

1. Sometimes higher-order diffraction is seen when the path difference is a multiple of λ .



X-Rays diffract treating inter atomic distances as slits.

For diffraction to occur, the condition given by **Bragg's Law** must be satisfied:

$$n\lambda = 2d \sin \theta$$

• λ can be roughly comparable to or smaller than $2d$, depending on the angle θ and the order n

- Wavelengths on the order of or smaller than $2d$ are suitable for diffraction studies because they allow constructive interference at observable angles.
- Typically, $\lambda \approx d$ or $\lambda \leq 2d$ is the practical working condition.

Why is Gama ray not used in getting Diffraction pattern from crystals?

Gamma rays are not used in X-ray diffraction (XRD) because their significantly shorter wavelength compared to X-rays results in much greater penetration power, making it difficult to achieve the necessary constructive interference patterns needed to analyze crystal structures; essentially, gamma rays would pass through a crystal without interacting with the atoms effectively, leading to a poor diffraction pattern

Key points about why gamma rays are not ideal for XRD:

- Wavelength mismatch:**

- The spacing between atoms in a crystal is on the order of the wavelength of X-rays, allowing for efficient diffraction, while gamma rays have a much shorter wavelength, meaning they interact poorly with the crystal lattice.

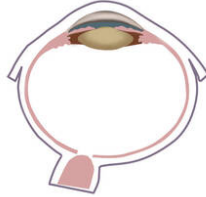
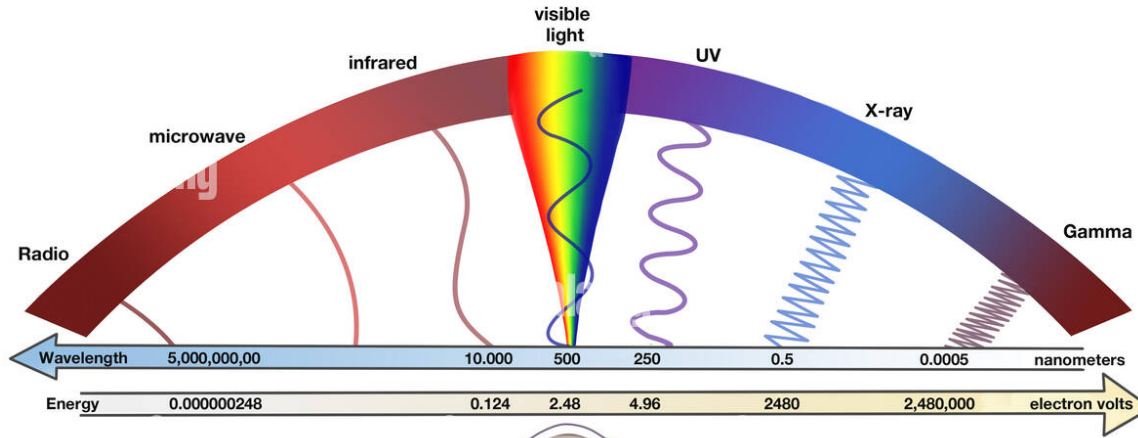
- High penetration power:**

- Due to their high energy, gamma rays easily penetrate through most materials, making it difficult to isolate the diffraction signal from a specific crystal plane.

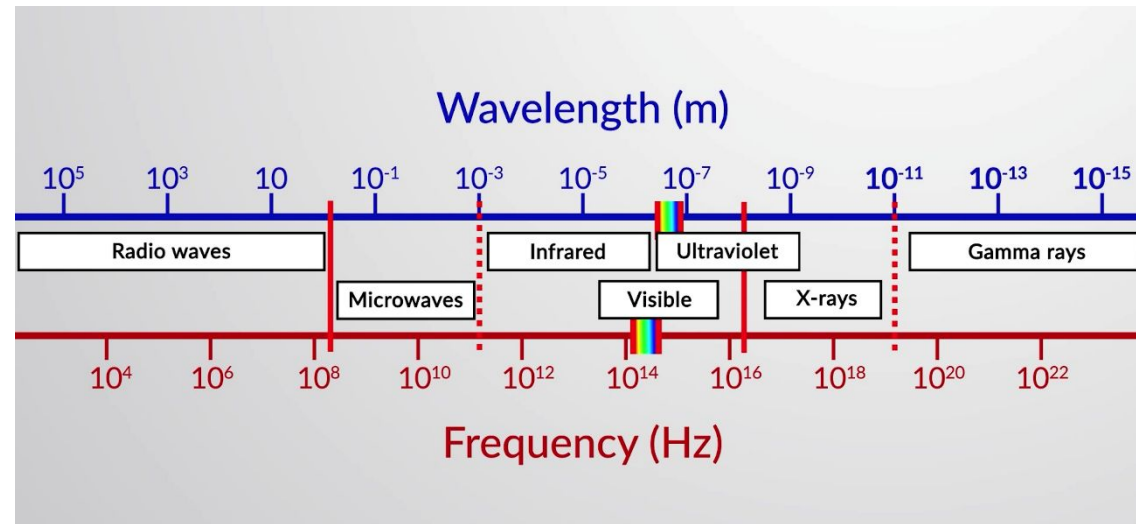
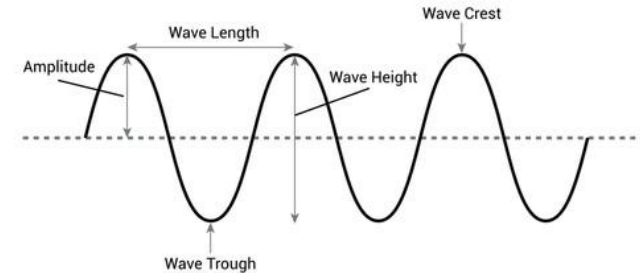
- Difficulties in generation and handling:**

- Producing a focused, controlled beam of gamma rays is considerably more challenging than generating X-rays, adding practical limitations to their use in XRD.

Idea about the wavelength of electromagnetic spectrum

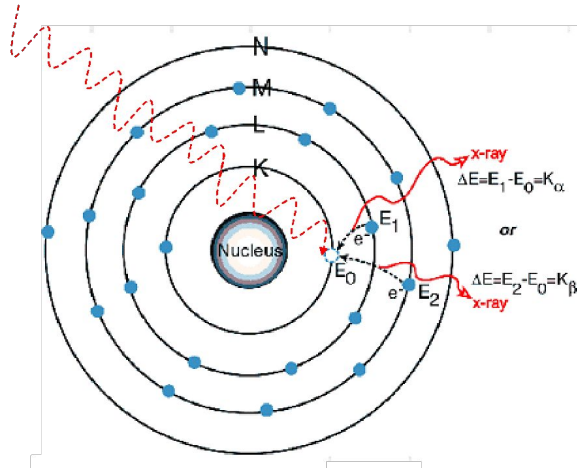


idea Parts of Wave



<https://www.alamy.com/stock-photo/electromagnetic-spectrum.html?cutout=1&sortBy=relevant>

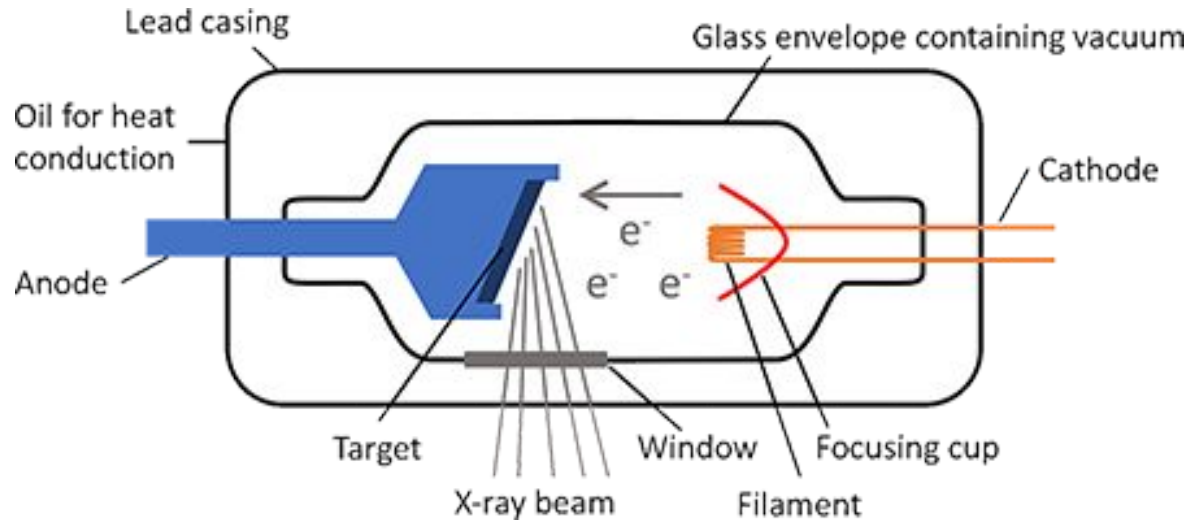
How is XRD produced?



An electron jumps from L or M shell to fill the vacancy, generating X-rays



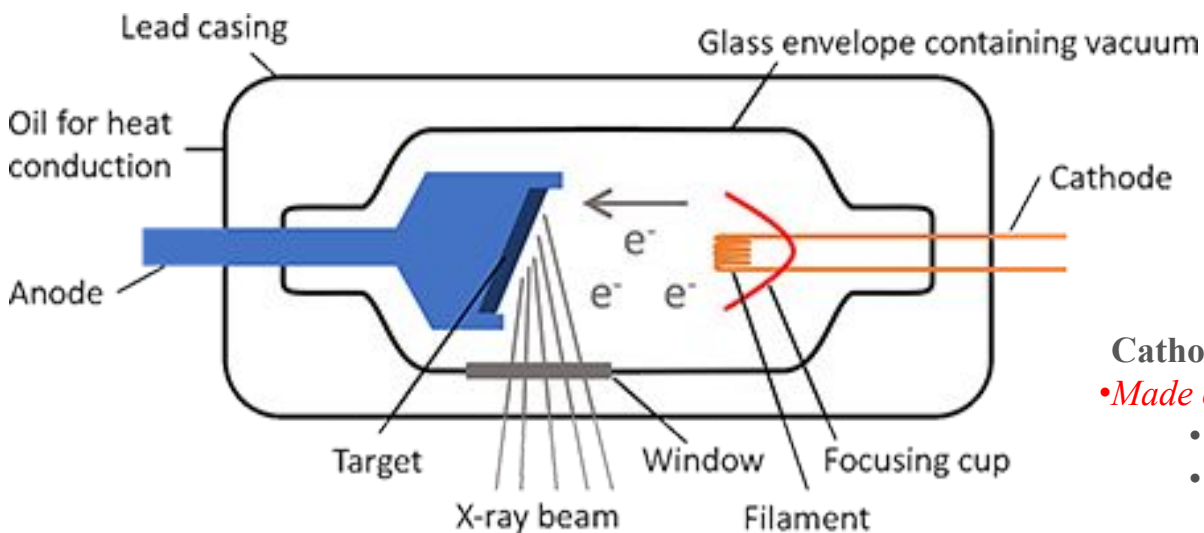
XRD-machine



X-ray production process

- A current is passed through the tungsten filament and heats it up.
- As it is heated up the increased energy enables electrons to be released from the filament through **thermionic emission**.
- The electrons are attracted towards the positively charged anode and hit the tungsten target with a maximum energy determined by the tube potential (voltage).
- As the electrons bombard the target they interact via Bremsstrahlung and characteristic interactions which result in the *conversion of energy into heat (99%) and x-ray photons (1%)*.
- *The x-ray photons are released in a beam with a range of energies (x-ray spectrum) out of the window of the tube and form the basis for x-ray image formation.*

Crucial components



Anode

- *Target made of tungsten for same reasons as for filament*
- Rhenium added to tungsten to prevent cracking of anode at high temperatures and usage
- Set into an anode disk of molybdenum with stem
- Positively charged to attract electrons
- Set at angle to direct x-ray photon beam down towards patient. Usual angle is $5^\circ - 15^\circ$

Apart from Tungsten, usually Copper metal is used as anode material.

Cathode: Filament

- *Made of thin (0.2 mm) tungsten wire because tungsten:*
 - has a high atomic number (A 184, Z 74)
 - is a good thermionic emitter (good at emitting electrons)
 - can be manufactured into a thin wire
 - has a very high melting temperature (3422°C)
- The size of the filament relates to the size of the focal spot. Some cathodes have two filaments for broad and fine focusing

Open the link to read more about XRD → <https://www.radiologycafe.com/frcr-physics-notes/x-ray-imaging/production-of-x-rays/>

Apart from Tungsten, usually Copper metal is used as anode material.

Why?

The wavelength of the emitted X ray depends upon the target material used.

$K\alpha$ (Å) for different Anode material. as follows.

Cr-2.29, Fe-1.94, Co-1.79, Cu-1.54, Mo-0.71, Ag-0.56

The rays emitted by copper anode is most suitable for diffraction to measure the lattice parameters in most of the cases.

In X-Ray diffraction, the X-rays are generated by hitting high energy ((45-50)kV & 40mA) and highly accelerated electrons produced from the cathode on a target material known as the anode. As such these highly energetic electrons transfer a lot of energy to the target material and heat it up. You can make a rough calculation that approximately 1.2kW of heat is generated in that process. This limits the choice for the target material to be of high melting point with excellent electrical conductivity. Only transition metals in the period table fulfill the above requirement. They are highly conductive and are having a high melting temperature. But for diffraction purposes, it's the wavelength of K-alpha which is used. The K-alpha values (Angstrom) for various transition metals as target materials are,

Cr Fe Co Cu Mo Ag

2.29 1.94 1.79 1.54 0.71 0.56

You can see here the wavelength of Mo and Ag are normally too short for most powder XRD. And if we use short-wavelength, they will scatter weakly and would shift the pattern towards lower Bragg angles which may result in loss of d-spacing accuracy. Moreover, the penetration depth would also be large.

For targets such as Cr and Fe, the wavelength is large which decreases the energy and intensity as well. Further, due to the higher wavelength, the XRD pattern shifts towards a larger Bragg angle which increases the acquisition time. As such due to low energy, the incoming rays may get absorbed by the specimen.

Thus, considering the above factors, it is optimized neither to have a high wavelength nor to have a short wavelength. That's why copper is being chosen as the target material. Its melting point is 1093 degrees and is highly conducting.

Again, as like mentioned by others, the fluorescence effect plays a role when a copper target is used for iron (Fe) and Manganese (Mn) containing materials. In order to avoid that, a Co target is used which has also a high melting temp with a K-alpha value of 1.79, pretty close to Copper.

Additional clarification:

*“For targets such as Cr and Fe, the wavelength is large which decreases the energy and intensity as well.” **How?***

Wavelength and energy relationship

The energy of X-rays is inversely proportional to their wavelength:

$$E = \frac{hc}{\lambda}$$

where:

- E is the photon energy,
- h is Planck's constant,
- c is the speed of light,
- λ is the wavelength.

So, **larger wavelength** → **smaller energy**

- Cr (Chromium) and Fe (Iron) have $K\alpha$ lines at longer wavelengths compared to heavier elements like Cu or Mo, which means their emitted X-rays have lower energy.

• Lower energy → reduced intensity

*Lower energy photons interact less effectively with the sample or detector, and are also more easily absorbed or scattered before detection, leading to **lower intensity**. Furthermore, lower-energy X-rays have weaker penetration and produce fewer counts during measurement, which translates into less intense signals.*

Additional clarification:

‘Further, due to the higher wavelength, the XRD pattern shifts towards a larger Bragg angle which increases the acquisition time.’ How?

Higher wavelength → larger Bragg angle

According to **Bragg’s law**:

$$n\lambda = 2d \sin \theta$$

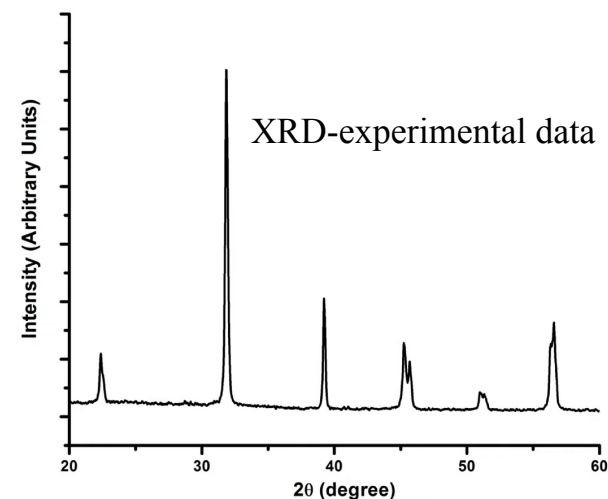
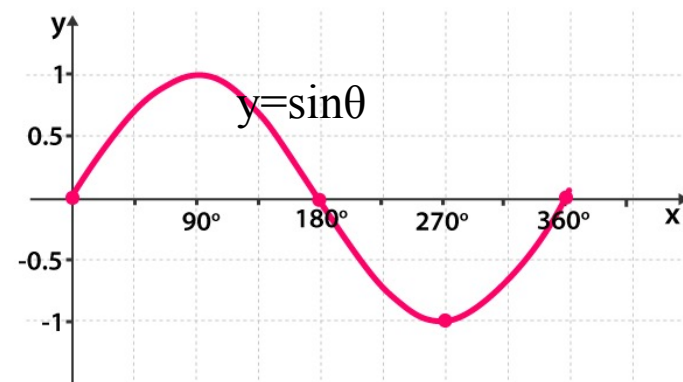
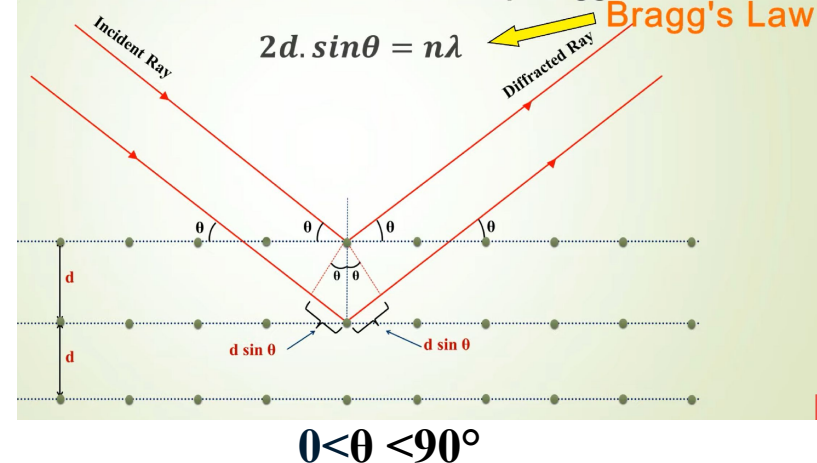
where:

- n is the order of reflection,
- λ is the wavelength,
- d is the spacing between planes,
- θ is the Bragg angle.

If λ increases, then for the same d , $\sin \theta$ must increase, meaning θ becomes larger.

- **Larger Bragg angle → increases acquisition time**

As the detector or diffractometer scans larger angles, the time taken to collect sufficient diffraction data increases because the scan covers a broader angular range and may require slower scan speeds to maintain resolution.

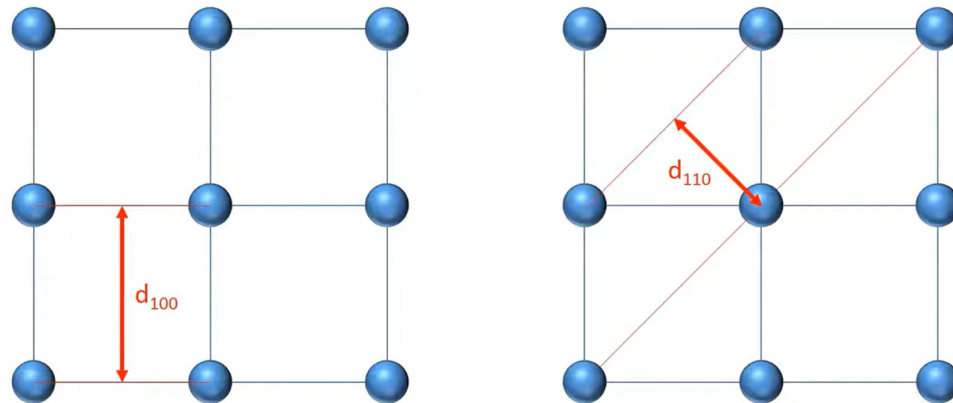


Peak positions

- The peak positions is determined by the inter-planar spacing for a family of planes

$$2d \cdot \sin\theta = n\lambda$$

Distance Between Planes

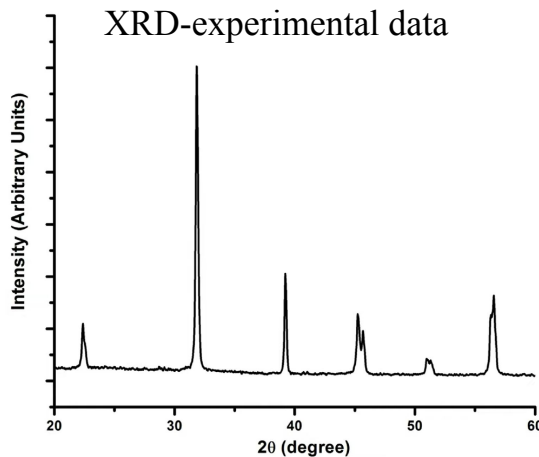


- A single crystal can diffract from multiple sets of planes (families of planes).**

Even though it's a single crystal, it has many atomic planes oriented in different directions.

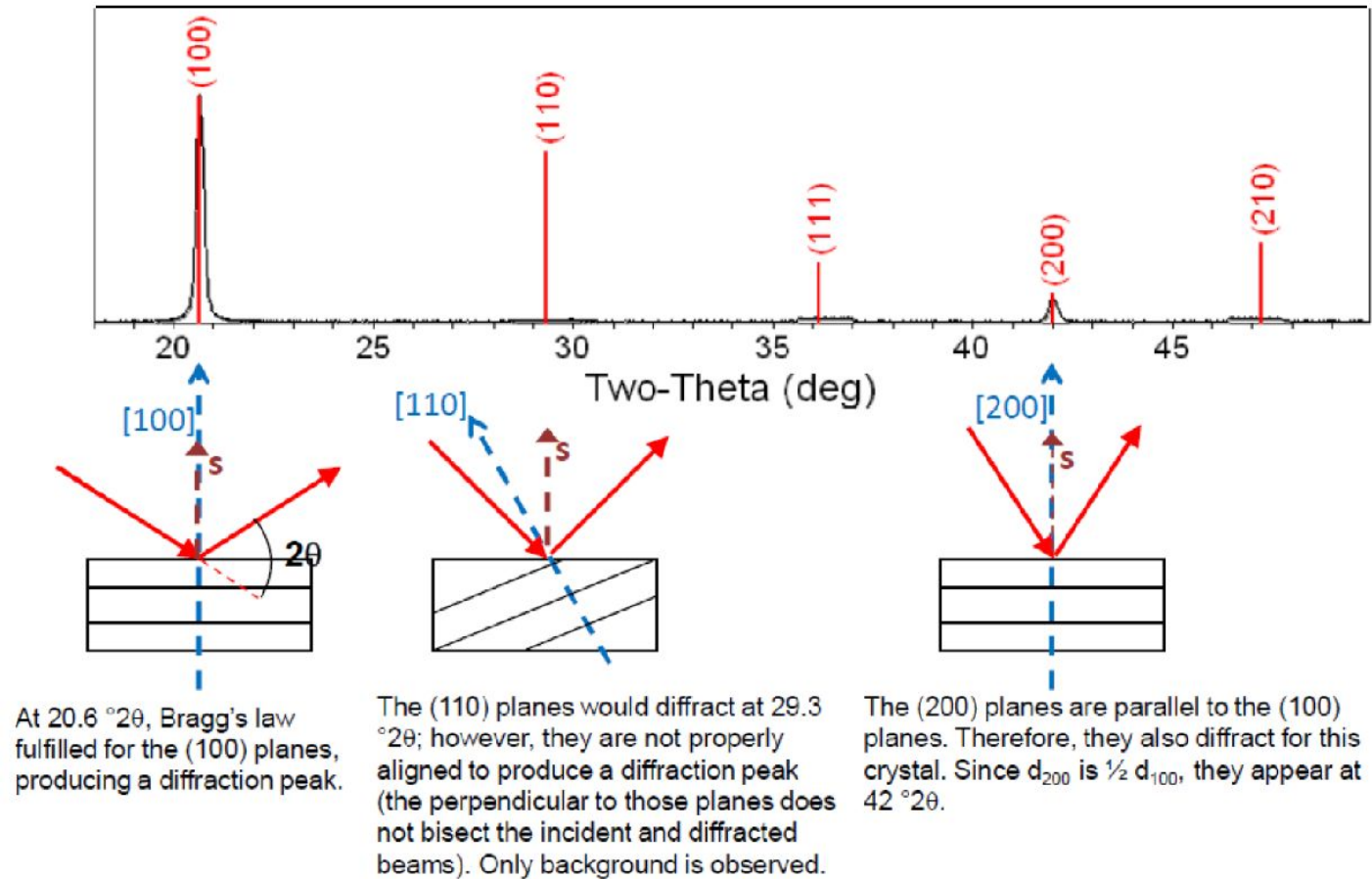
- Multiple diffraction peaks appear, each corresponding to a specific set of planes that satisfy Bragg's Law at certain angles.*

- In a Bragg-Brentano geometry**, the crystal is rotated, and diffraction is measured over a range of angles θ , so peaks from several planes are observed



Peak Intensity

A single crystal specimen in a Bragg-Brentano diffractometer would produce only one family of peaks in the diffraction pattern



Why different planes give different intensities for the same material.

1. Structure Factor Differences

- The structure factor $F(hkl)$ depends on the positions and types of atoms within the unit cell.
- For some planes, atoms are aligned in ways that cause **constructive interference**, resulting in a strong signal.
- For others, the interference may be partial or even destructive, leading to weak or absent peaks.
- Even if the material is the same, different planes have different atomic arrangements, so the intensity changes.

2. Multiplicity Factor

- Some planes have more equivalent orientations in the crystal.
- For example, the $\{111\}$ family might have more symmetrically equivalent planes than $\{200\}$, so the $\{111\}$ peak may appear stronger.
- This is purely a geometric effect.

3. Interplanar Spacing and Bragg's Condition

- Different planes have different d-spacing.
- The angle θ at which diffraction occurs changes, affecting the geometric and polarization factors.
- At higher angles, intensities can drop due to instrumental and sample factors.*

4. Preferred Orientation (Texture)

- Even in the same material, if crystals are not randomly oriented but have preferred orientations, certain planes will diffract more strongly.
- For example, thin films often show strong peaks from planes parallel to the film surface.

5. Absorption and Thickness Effects

- Different planes scatter X-rays at different angles, leading to variations in how much material the beam passes through before detection. Some reflections are more absorbed, weakening their intensity.

6. Thermal Effects

- Vibrations of atoms (thermal motion) can weaken reflections from certain planes more than others, especially those at higher angles.

Peak Width

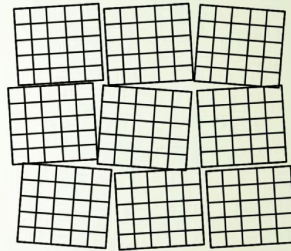
Width of a peak basically depends on two factors-

1. Crystallite size
2. Lattice Strain

Crystallite size- it is the size of coherently diffracting piece of a crystal. In an XRD pattern it can be calculated by Sherrer's formula.

Crystallite
Size

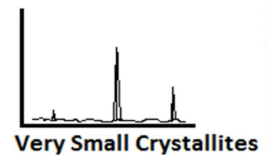
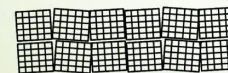
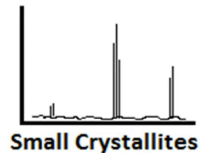
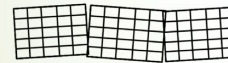
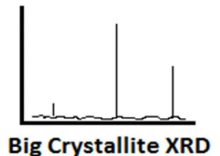
$$d = \frac{K\lambda}{\beta \cos\theta}$$



Here K is a constant generally taken as 0.9 .. λ is the wavelength used. B is the FWHM of a peak and θ is the half of corresponding peak angle.

Crystallite size refers to the size of a coherent, crystalline domain within a material — that is, a region where the crystal lattice is continuous and without defects or grain boundaries that disrupt the periodic arrangement of atoms. It's important not to confuse **crystallite size** with **particle size** — a particle may consist of many crystallites.

Peak Width : Crystallite Size

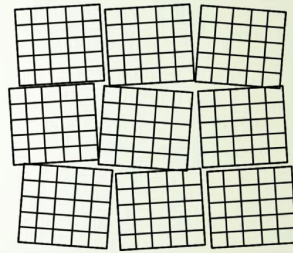


Peak Width

1. Crystallite size

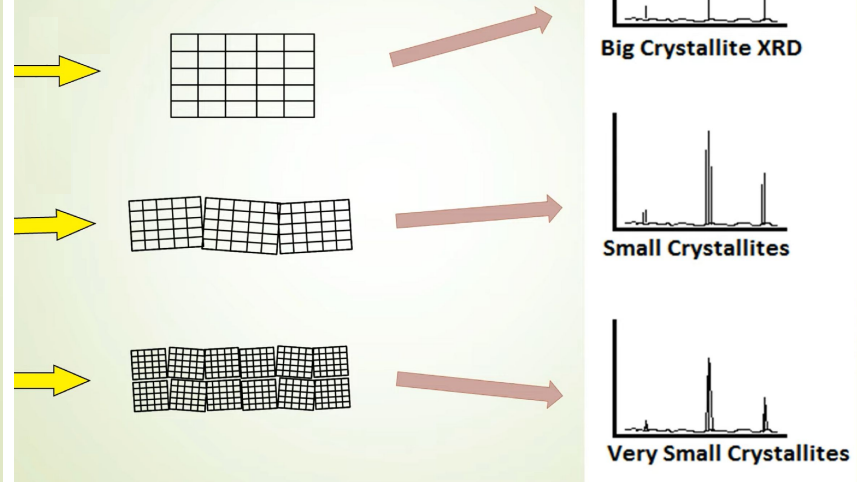
Crystallite size- it is the size of coherently diffracting piece of a crystal. In an XRD pattern it can be calculated by Sherrer's formula.

$$\rightarrow d = \frac{K\lambda}{\beta \cos\theta}$$



Here K is a constant generally taken as 0.9 .. λ is the wavelength used. B is the FWHM of a peak and θ is the half of corresponding peak angle.

Peak Width : Crystallite Size

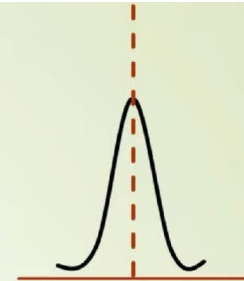
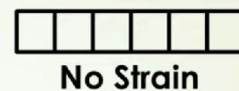


Peak Width

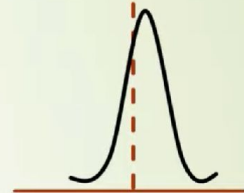
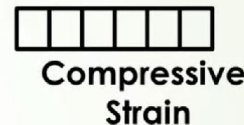
2. Lattice Strain

Peak Width : Strain

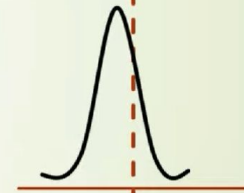
• **Case 1 → No Strain →** Peak at its natural position.



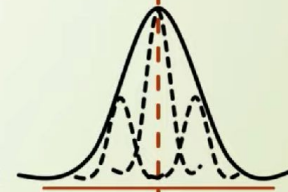
• **Case 2 → Compressive Strain →** Lattice size decreases 2θ increases.



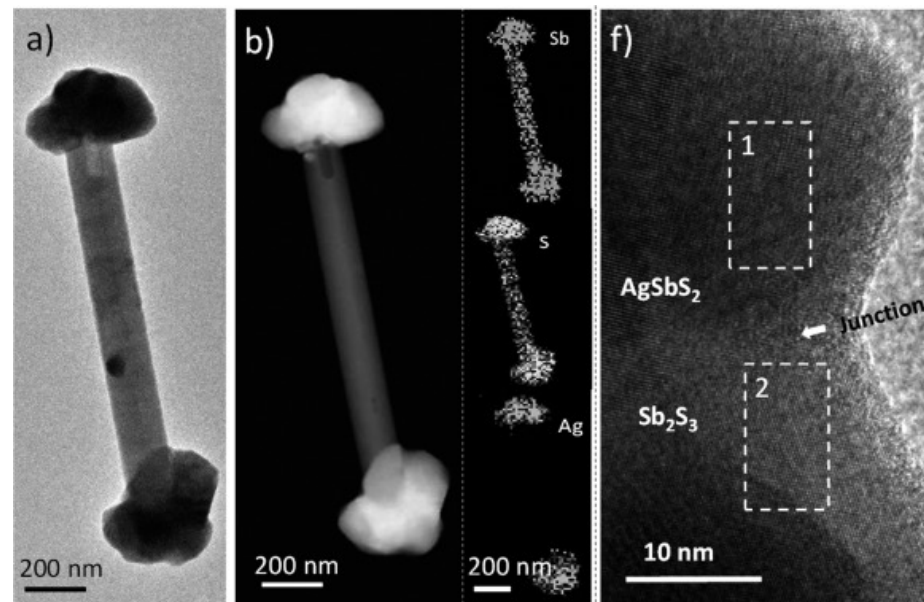
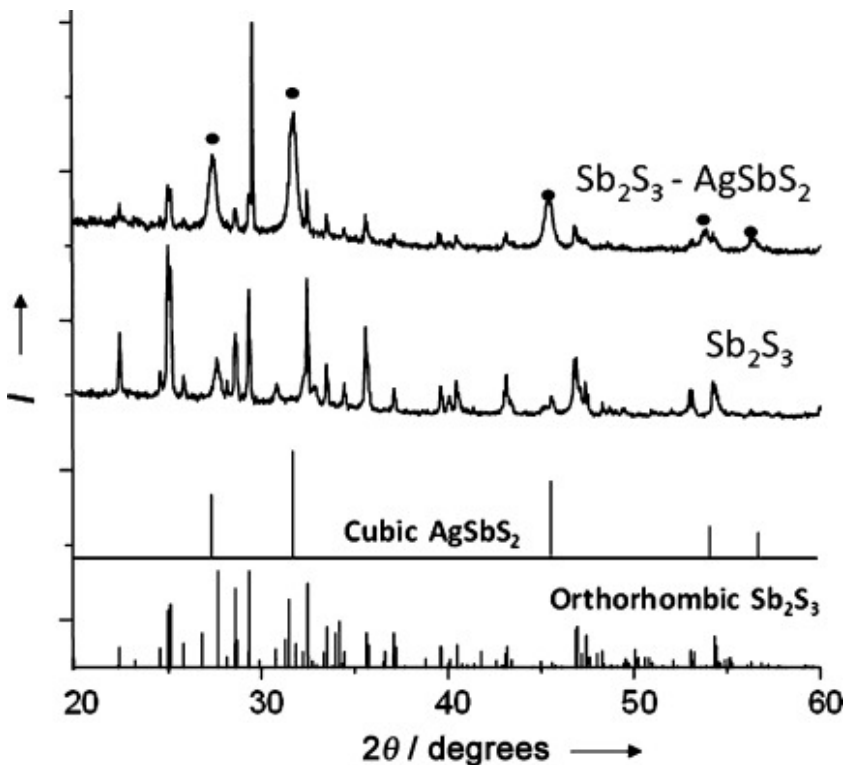
• **Case 3 → Tensile Strain →** Lattice size increases 2θ decreases.



• **Case 4 → Non-uniform Strain →** Peak gets broader.



XRD analysis from a laboratory synthesized sample



XRD analysis from a laboratory synthesized sample

Folded Nanosheets: A New Mechanism for Nanodisk Formation

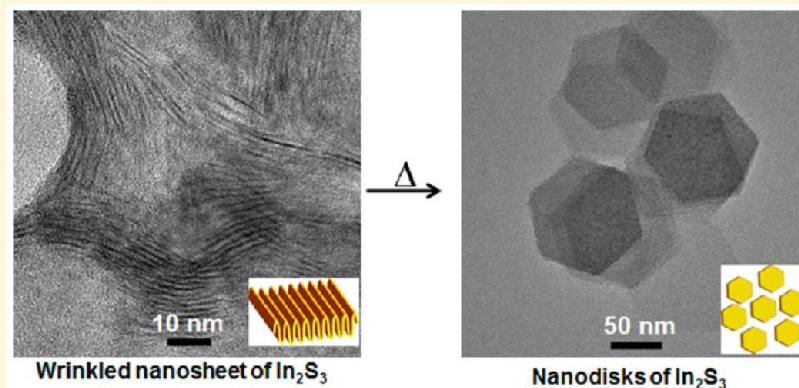
Suresh Sarkar,[†] Alice D. P. Leach,[†] and Janet E. Macdonald^{*,†,‡}

[†]Department of Chemistry, and [‡]Vanderbilt Institute of Nanoscale Science and Engineering, Vanderbilt University, Nashville, Tennessee 37235, United States

Supporting Information

ABSTRACT: Two-dimensional (2D) nanostructures have generated significant interest in the scientific community as a result of their high surface area and remarkable optoelectronic properties. In this work, we present a simple colloidal synthesis for ultrathin nanosheets of both In_2S_3 and CuInS_2 . Contradictory phase designations exist for In_2S_3 at the nanoscale as a result of overlapping reflections from multiple phases in X-ray diffraction. We use high-resolution transmission electron microscopy to demonstrate definitively that hexagonal $\gamma\text{-In}_2\text{S}_3$ is formed. Continued heating of these nanostructures results in the formation of nanodisks twice the thickness of the original structure via folding and tearing of the intermediate sheet.

Treatment of the In_2S_3 nanosheet with Cu^+ accelerates this process of tearing of the nanosheet of In_2S_3 and results in the formation of one-dimensional nanoribbons of CuInS_2 . High-resolution transmission electron microscopy images and medium-angle X-ray scattering measurements indicate that these nanoribbons consist of face-to-face stacked nanodisks.



XRD analysis from a laboratory synthesized sample

Sometime the real experimental sample gives very complex XRD pattern

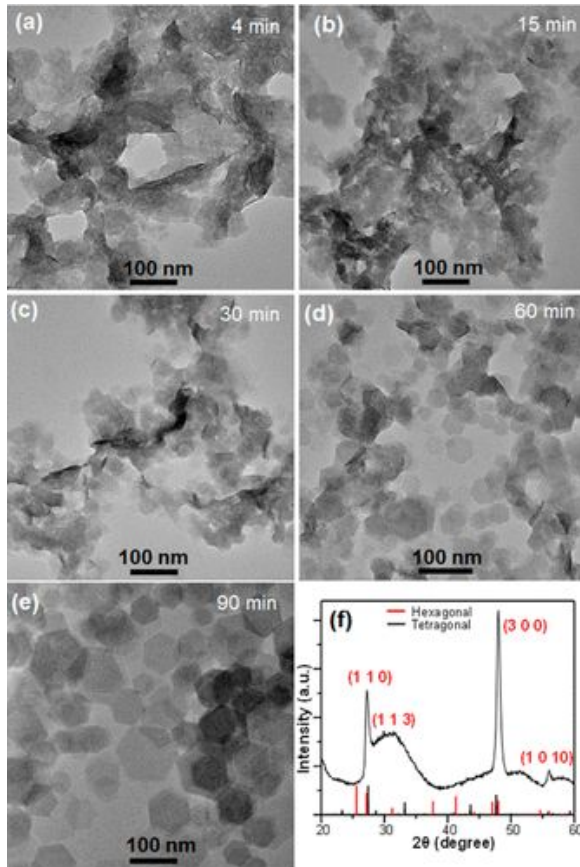


Figure 1. (a-e) Shape transformation of 2D nanosheets to nanodisks of In_2S_3 with annealing of the reaction mixture at 215 °C. (f) Powder XRD of In_2S_3 nanosheets at 4 min with expected reflections for tetragonal $\beta\text{-In}_2\text{S}_3$ (JCPDS 25-0390) and hexagonal $\gamma\text{-In}_2\text{S}_3$ (JCPDS 33-0624) shown below. The experimental reflections are assigned to the appropriate Miller indices of the hexagonal phase.

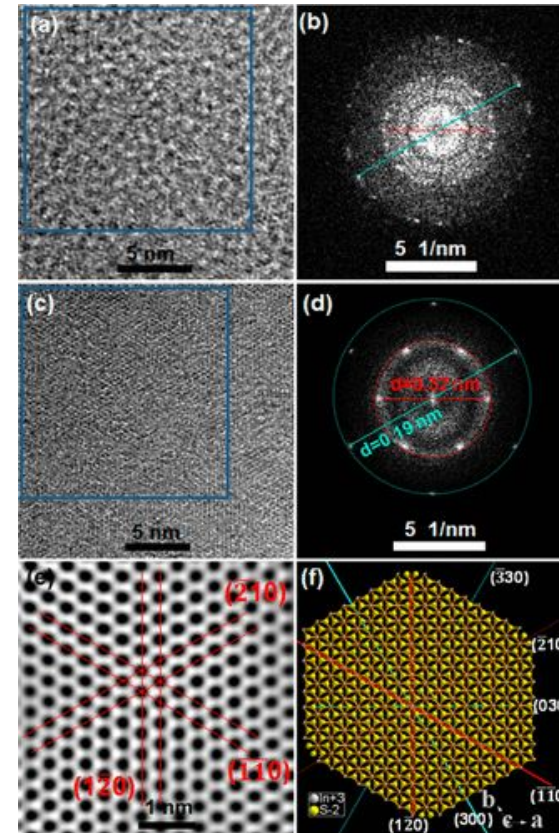
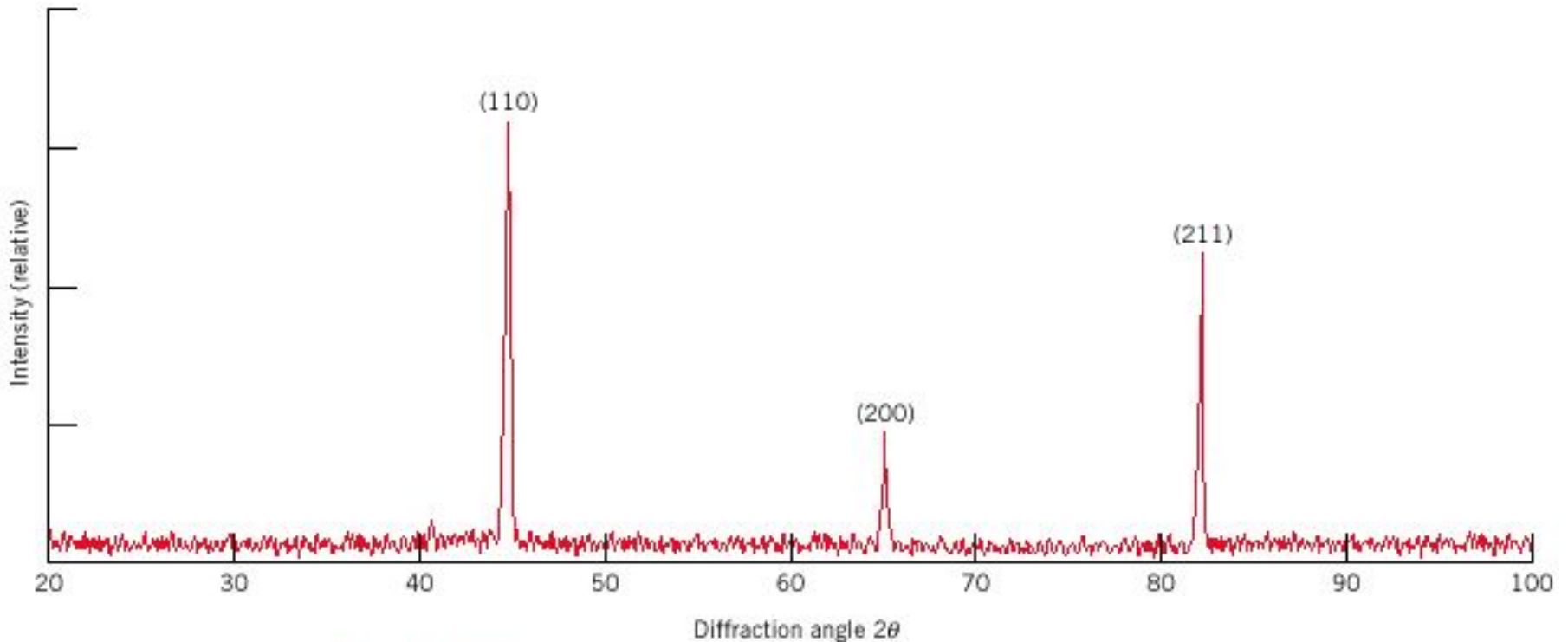


Figure 2. (a) HR-TEM image of an In_2S_3 nanosheet in a flat area. (b) FFT of the selected area in panel a. (c) HR-TEM image of an In_2S_3 nanodisk. (d) FFT of the selected area in panel c. (e) Simulated HR-TEM for the selected area in panel c. (f) Atomic model correlating the observed d spacing to atomic planes for the simulated HR-TEM in panel e.

Below is the typical diffraction pattern for polycrystalline α -iron (BCC)



Draw the unit cells indicating planes you see in the above XRD (three unit cells for three different planes) and calculate the d-spacing of all the peaks (given 2θ values for (111), (200), (211) are 45, 65 and 82 respectively).

What is X-ray Diffraction and what are its application in chemistry?

Ans: The phenomena by which X-rays are reflected from the atoms in a crystalline solid is called diffraction. The diffracted X-rays generate a pattern that reveals structural orientation of each atom in a given compound. X-ray diffraction is extensively used in chemistry for the characterization of organic and inorganic compounds that are made for pharmaceutical companies or making batteries of the cell phones.

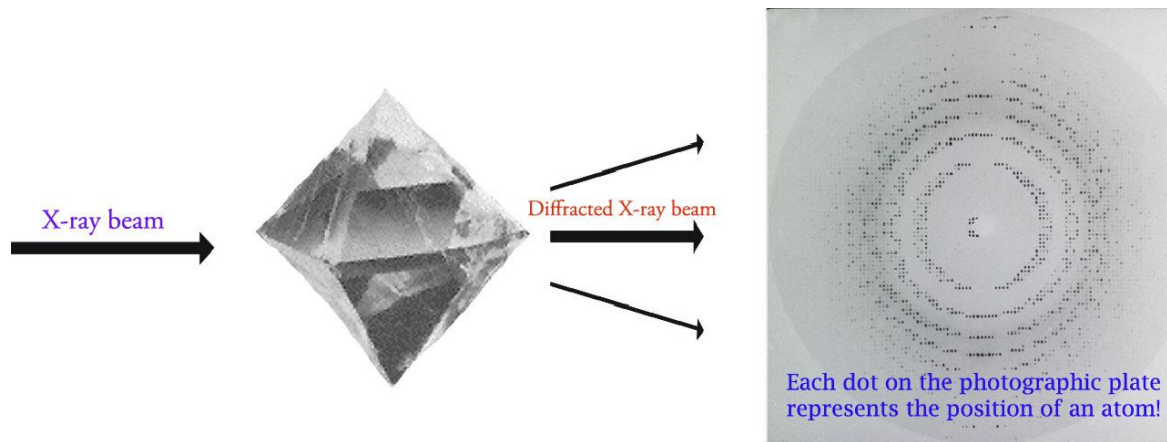
XRD finds the geometry or shape of a molecule using X-rays. This technique is based on the elastic scattering of X-rays from structures that have long range order (crystalline solids). XRD technique is divided into two categories based on the morphology and size of sample:

1. If a crystal sample is large enough then it can be analyzed using **X-ray Single Crystal diffraction**, which solves for the complete structure ranging from simple inorganic solids to complex macromolecules. When we say “crystals of large enough size” please note that they are still small enough for the normal eyes (most of the time like a speck of dirt!).

2. Does not form crystals large enough, then the sample is analyzed by using **X-ray Powder diffraction** (XRPD) technique. Powders of crystalline materials diffract X-rays. A beam of X-rays passing through a sample with randomly-oriented micro crystals produces a pattern of rings on a distant screen. XRPD provides less information than X-ray single crystal diffraction; however, it is much simple and faster. XRPD is useful for confirming the identity of a solid material and determining crystallinity and phase purity.

What is the principle of XRD?

Ans: XRD finds the geometry or shape of a molecule using X-rays. XRD techniques are based on the elastic scattering of X-rays from structures that have long range order. The X-rays get diffracted by a crystal because the wavelength of X-rays is similar to the inter-atomic spacing in the crystals.



When the X-ray beam encounters the regular three-dimensional arrangements of atoms in a crystal, most of the X-rays will destructively interfere with each-other and cancel each-other out, but in some specific directions the X-ray beams interfere constructively and reinforce one another. It is these reinforced diffracted X-rays that produce the characteristic X-ray diffraction pattern that is used for crystal structure determination.

W.L. Bragg in the early 19th century showed that diffracted X-rays act as if they were 'reflected' from a family of planes within crystals. Later named after him, these Bragg's planes are the rows of atoms that make up the crystal structure as shown in the figure below.

Does the XRD help in determining the crystal structure and molecular formula of a crystalline compound?

Ans: XRD is an important method to characterize the structure of crystalline material. It can be used to determine either the lattice parameters, arrangement of individual atoms in a single crystal, or the phase analysis in case of polycrystalline materials and compounds. With the knowledge of XRD and *crystallography*, it is possible to determine the crystal structure and molecular formula of a crystalline compound

What is X-ray Crystallography?

Ans: X-ray crystallography is a method of determining the arrangement of atoms within in a crystal, in which a beam of X-rays strikes a crystal and scatters into many different directions. From the angles and intensities of these scattered beams a three-dimensional picture of the density of electrons within the crystal can be produced. From this electron density, the mean positions of the atoms in the crystal can be determined, as well as their chemical bonds, their disorder and various other information. Since very many materials can form crystals - such as salts, metals, minerals, semiconductors, and various inorganic, organic and biological molecules. X-ray crystallography has been fundamental in the development of many scientific fields.

What is a Crystal Structure?

Ans: A crystal structure is a unique arrangement of atoms/ molecules/ ions in a crystal. A crystal structure is composed of a motif, a set of atoms/ molecules/ ions arranged in a particular way, and a lattice. Motifs are located upon the points of a lattice, which is an array of points repeating periodically in three dimensions. The points can be thought of as forming identical tiny boxes, called unit cells, that fill the space of the lattice. The lengths of the edges of a unit cell and the angles between them are called the lattice parameters